

DAINS Project Report

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1 Introduction

Computer simulations (FEA) can model metal bending and stress very well, but running them over and over takes too much time. This report combines computer simulations, active learning, and deep learning to speed up the process while keeping high accuracy.

2 Physics-based simulations

2.1 Methods

Load-controlled bending simulation was performed on a square specimen utilized a bilinear elastoplastic model with kinematic hardening. Equilibrium was resolved via a modified „Initial Stiffness“ Newton-Raphson scheme ($\mathbf{K}_t = \text{const}$) (1) [1] paired with radial-return mapping for von Mises stress integration (2) [2].

$$\mathbf{K}_t \Delta \mathbf{u} = \mathbf{f}_{t+\Delta t}^{ext} - \mathbf{f}_t \quad (1)$$

$$\mathbf{s}_{n+1}^T = \mathbf{s}_n + 2G\Delta \mathbf{e}_{n+1}, \mathbf{s}_{n+1} \equiv P(\mathbf{s}_{n+1}^T) \quad (2)$$

Pseudo-time steps are solved iteratively until residual tolerances or maximum iteration limits are met, updating internal stresses and plastic history variables at each step. The simulation configuration, finite element, and boundary condition are detailed below (see Table 1).

Table 1. Details of simulation configuration, finite element, and boundary condition

Parameter	Value	Boundary condition
Domain geometry	1 m $\times$ 1 m square (1 m thickness)	Linear bending profile
Mesh discretization	26 $\times$ 26 quadratic element	$u_x(y) = \pm u_0 \cdot \frac{2(y-\frac{H}{2})}{H}$ on left/right domain boundaries.
Elastic constant (E, ν)	70 GPa, 0.3	
Plastic parameters (σ_0 , E_t)	250 MPa, 0 Pa (pure elastoplastic)	
Max. boundary displacement (u_0)	3.5 mm over 5 load steps	
Newton-Raphson Setup	$\epsilon_{\text{res}} = 10^{-10}$, max. 100 iterations	

2.2 Results

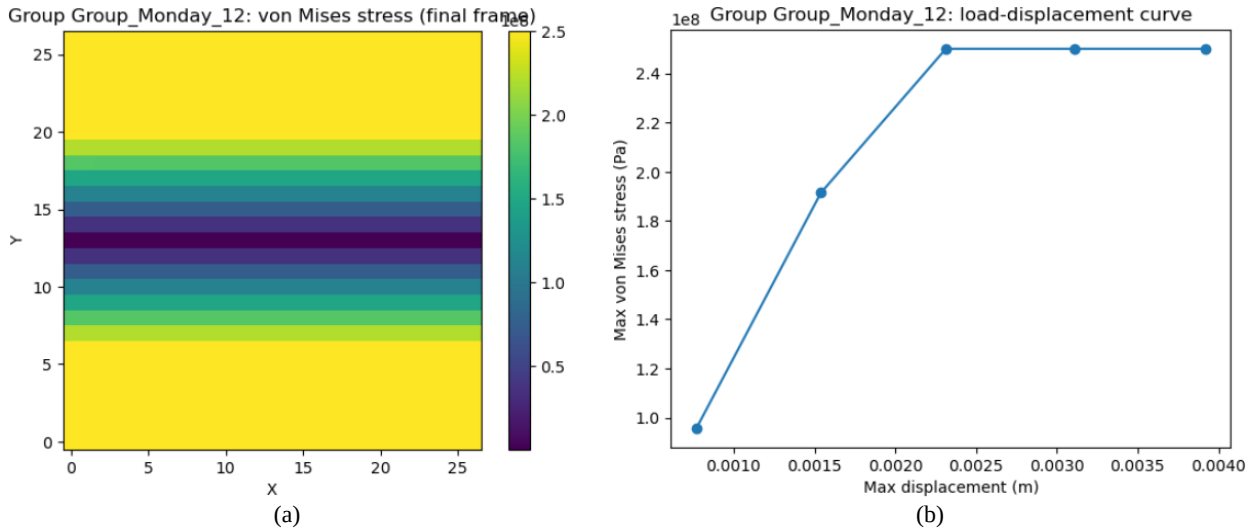


Figure 1. Result of physics-based simulation: (a) von Mises stress field, (b) stress-displacement curve

Von Mises stress maps (Figure 1a) and stress-displacement curves (Figure 1b) show pure bending behavior: peak compression and tension develop at the top and bottom edges, while the neutral axis remain stress-free. Maximum von Mises stress reaches the initial yield stress ($\sigma_0 = 250$ MPa) and caps there because $E_t \approx 0$ as plastic flow develops along the outer boundaries.

3 Active learning

3.1 Methods

Using the Project 1 FE model as a black box, multi-objective optimization maximized maximum von Mises stress (max_mises) and minimized maximum final displacement (max_u) across $E \in [50, 90]$ GPa and $u_0 \in [1, 5]$ mm. Executed over 20 Ax active learning trials, space-filling Sobol sampling preceded Gaussian Process modeling (2). The GP prior and dataset induce a posterior $f(x)$, while acquisition function (3) determines the next evaluation point in X via proxy optimization (4) [3].

$$\{x_n, y_n\}_{n=1}^N, \text{ where } y_n \sim \mathcal{N}(\mu, \sigma^2) \quad (2)$$

$$a : \mathcal{X} \rightarrow \mathbb{R}^+ \quad (3)$$

$$x_{next} = \operatorname{argmax}_x a(x) \quad (4)$$

3.2 Results

While u_0 directly governs max_u (Figure 2a), max_mises depends on both E and u_0 , reaching the 250 MPa yield plateau under high stiffness and moderate-to-high u_0 (Figure 2b). The Pareto front highlights an active trade-off range (max_u $\in [1.1, 1.7]$ mm), where u_0 sharply increases stress (Figure 3). Past max_u ≈ 1.7 mm, full yielding shifts deformation into pure plastic flow without stress gain, leaving points like (1.8 mm, 160 MPa) inside the sub-optimal region.

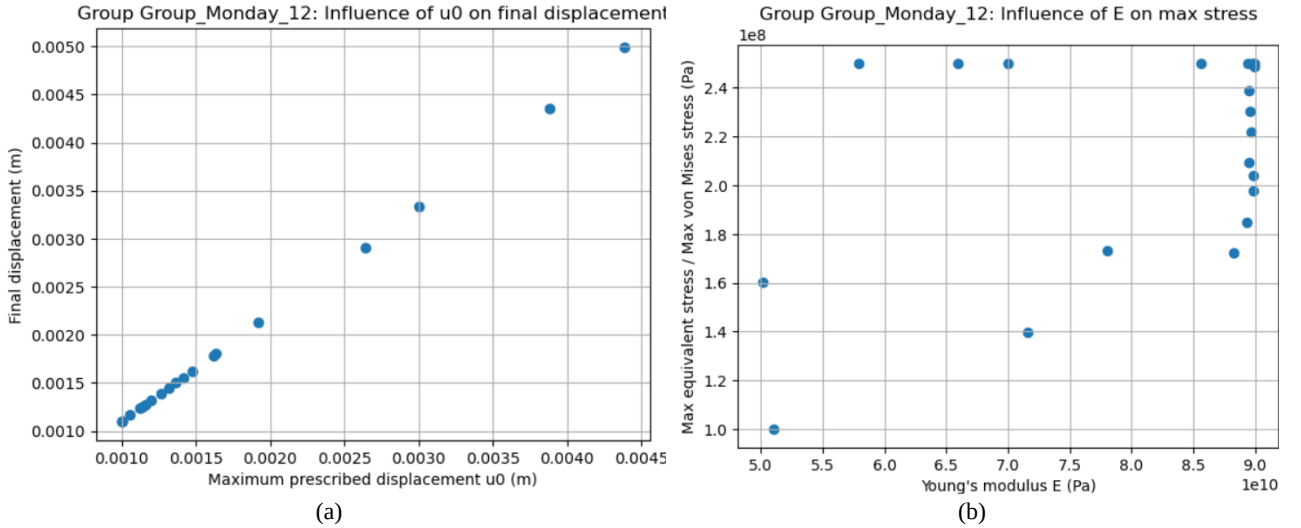


Figure 2. Results of active learning: (a) u_0 vs. max_u, (b) E vs. max_mises

4 Deep learning

4.1 Methods

Two surrogate models – a DNN and a 2D CNN – were trained on preprocessed (shuffled, split, normalized) elastoplastic FEA data from Projects 1 and 2. The DNN maps scalar inputs max_u, E , and σ_0 to scalar outputs maximum force (max_f) and max_mises. Meanwhile, the CNN predicts the

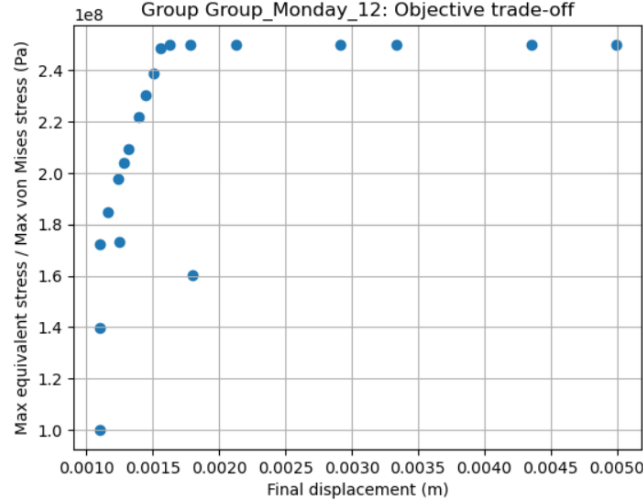


Figure 3. Result of active learning: Pareto front plot

full von Mises stress field (mises) given boundary displacement (U_{boundary}), E , and σ_0 .

Implemented in TensorFlow/Keras, both models use time-distributed surrogate models weight sharing across time steps and were trained with Adam ($\alpha = 10^{-3}$), MSE loss, and MAE metrics. The sequence-based DNN contains three hidden layers (64, 64, 32 ReLU units) with a linear output layer, trained for 300 epochs (batch size 2, 25% validation split). The 2D CNN features four 32-filter (7×7) layers and one 16-filter (5×5) layer with ReLU and same padding, ending in a 1×1 convolutional layer. It was trained for 200 epochs (batch size 8, 20% validation split).

4.2 Results

Evaluation of training loss curves was limited by the missing DNN loss graph and non-logarithmic scaling (Figure 4). However, closely aligned final training and validation MSE losses - reaching 6.2786×10^{-4} vs. 3.4962×10^{-4} for the DNN at 300 epochs, and 3.298×10^{-4} vs. 4.0502×10^{-4} for the CNN at 200 epochs - indicate no significant overfitting (Table 2). This is further supported by near-perfect predictive accuracy ($R^2 = 0.9995$) across all targets (max_f, max_mises, and spatial mises fields; Table 2, Figure 5-6).

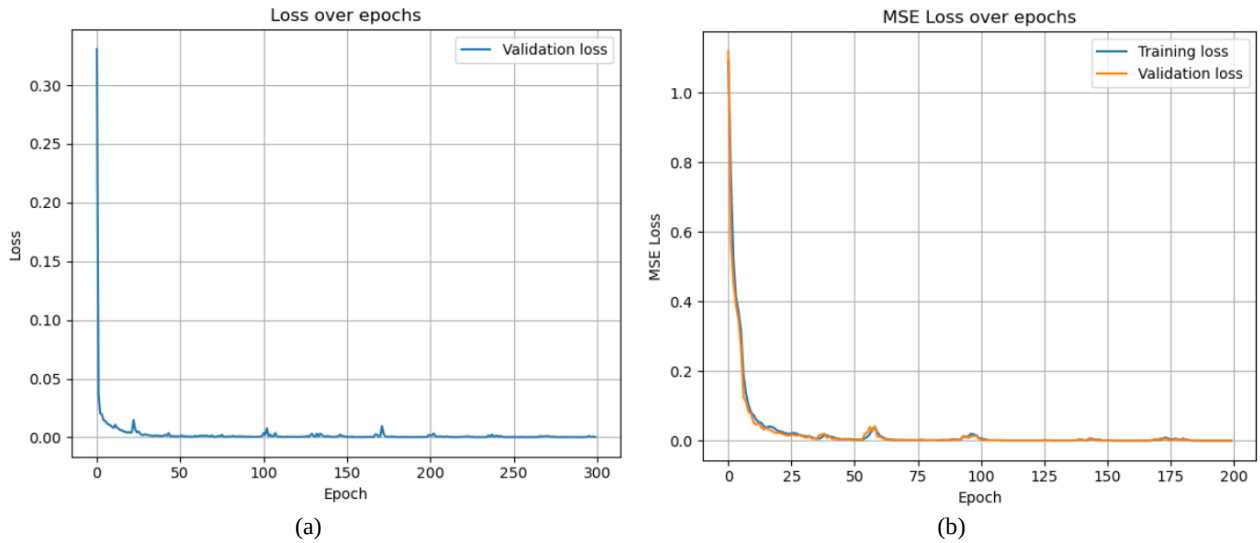
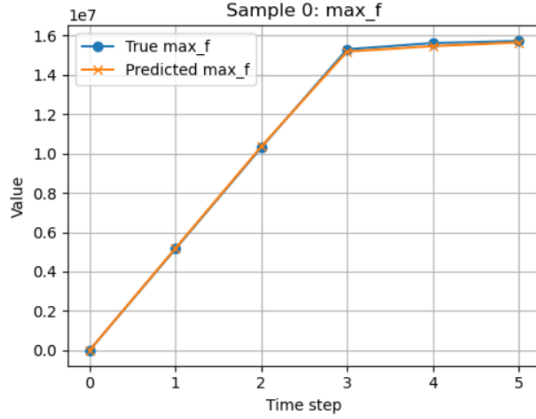


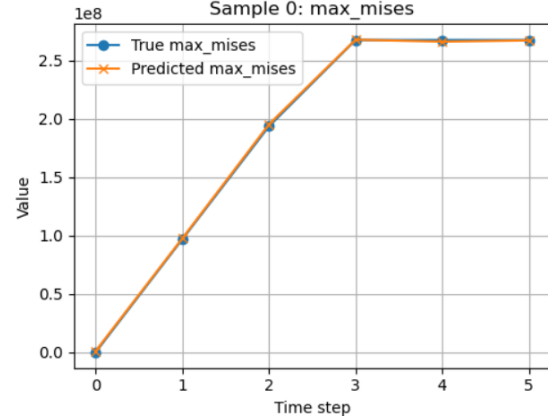
Figure 4. Neural networks' loss performance: (a) DNN loss curve, (b) CNN loss curves

Table 2. MSE loss

DNN			CNN		
Epoch	Training loss	Validation loss	Epoch	Training loss	Validation loss
298/300	2.7671×10^{-4}	4.9839×10^{-4}	198/200	4.5290×10^{-4}	5.5734×10^{-4}
299/300	2.3090×10^{-4}	5.5862×10^{-4}	199/200	4.4027×10^{-4}	3.8269×10^{-4}
300/300	6.2786×10^{-4}	3.4962×10^{-4}	200/200	3.2980×10^{-4}	4.0502×10^{-4}



(a)



(b)

Figure 5. Results of DNN: (a) max_f output, (b) max_mises output

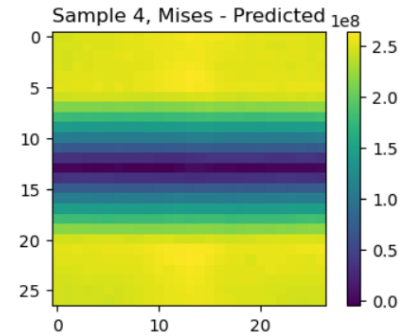
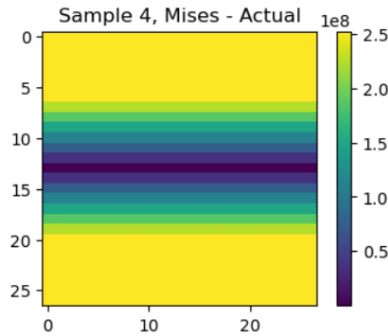


Figure 6. Results of CNN: Visualization of actual and predicted von Mises stress field

5 Conclusion

The computer simulation successfully showed how the metal bends and stretches up to its limit 250 MPa. Active learning then found the best balance between stress and displacement in just 20 tries, and the neural networks learned to predict the full stress patterns almost perfectly ($R^2 = 0.9995$).

Reference

1. Owen, D.R.J., Gomez, C.M.B.: An appraisal of numerical solution techniques for elasto-plastic and elasto-viscoplastic material problems. *Convergence* 1(1), (1981).
2. Simo, J.C., Taylor, R.L.: Consistent tangent operators for rate-independent elastoplasticity. *Computer Methods in Applied Mechanics and Engineering* 48(1), 101-118 (1985).
3. Snoek, J., Larochelle, H., Adams, R.P.: Practical Bayesian optimization of machine learning algorithms. *Advance in Neural Information Processing Systems* 25, 2960-2968 (2012).